

EXAM POV NOTES (Q&A FORMAT)**10 Marks Questions**

1. Compare and contrast contingency planning versus replanning approaches in non-deterministic domains. In the context of a self-driving car navigating rush hour, analyze the effectiveness of both methods concerning factors such as computational resources, adaptability, and reliability.

Non-deterministic domains, characterized by uncertain action effects and dynamic environments, necessitate advanced planning approaches like contingency planning (pre-calculating branches) and replanning (re-solving upon failure) to manage uncertainty.

- **Contingency Planning:** This approach involves creating a comprehensive, branching plan structure that explicitly accounts for all different possible outcomes of uncertain actions.
- **Replanning:** This method starts with a simple initial plan and triggers a completely new planning process from the current state only if the initial plan fails during execution.
- **Computational Resources (Contingency):** Contingency planning can be highly complex and computationally demanding for large problems because it must explicitly map out all possible futures and plan for them in advance.
- **Computational Resources (Replanning):** Replanning can be computationally expensive if failures are frequent, necessitating repeated calls to the planning engine, though the initial planning phase is simple.
- **Adaptability:** Replanning is highly adaptable to entirely unexpected changes in dynamic environments, as it restarts the problem-solving process based on the new reality (e.g., handling a surprise road closure).
- **Reliability (Contingency):** Contingency planning offers strong reliability regarding anticipated uncertainties, as the response to known variations is pre-baked into the plan structure.
- **Reliability (Replanning):** Reliability relies on the system's ability to detect failures and quickly generate a valid new plan within the execution timeframe.
- **Car Example (Contingency):** A car might have a contingency plan for a traffic light turning yellow: "If light is yellow, then brake; if light is green, then proceed".
- **Car Example (Replanning):** If a self-driving car's initial route plan fails because of an accident or unexpected roadblock, it halts execution and computes an entirely new route from the current location (e.g., initial route $A \rightarrow B \rightarrow C$ fails at $A \rightarrow B$, so it plans $A \rightarrow D \rightarrow C$).
- **Effectiveness Summary (Rush Hour):** For rush hour, Replanning is often crucial for factors like adaptability to sudden, unexpected changes (accidents, unpredictable human drivers), while Contingency Planning is better suited for handling known, probabilistic action outcomes (e.g., minor sensor noise).

2. Explain the steps of the Partial-Order Planning (POP) algorithm. Detail the critical concepts of a causal link and a threat, and describe how POP resolves flaws and threats during the development of a partial plan.

Partial-Order Planning (POP) is the main algorithm for Plan-Space Search, which iteratively searches through the space of partial plans, starting with an empty plan and refining it by adding constraints and resolving identified flaws.

- **Initialization:** The algorithm starts by establishing two sentinel actions representing the boundary conditions: the initial state action (Start) and the goal state action (Finish).

- **Partial Plan Focus:** POP works by maintaining a partial plan, which is a set of actions connected by ordering constraints, focusing on achieving the goal without specifying a strict linear sequence initially.
- **Flaw Identification:** In each iteration, the algorithm chooses an unsatisfied precondition, which is referred to as a "flaw," existing within the current partial plan.
- **Flaw Resolution Method 1 (Existing Action):** A flaw can be resolved by integrating an existing action already in the plan whose effect successfully achieves the necessary precondition.
- **Flaw Resolution Method 2 (New Action):** Alternatively, a new action can be added to the plan specifically to achieve the precondition needed to resolve the flaw.
- **Causal Link Definition:** A causal link is a formal concept that explicitly shows how the effect of one action serves to fulfill the precondition of a subsequent action.
- **Causal Link Addition:** Once a flaw is resolved by an action, a causal link is added, indicating that the new action provides the necessary state for the subsequent action, alongside new ordering constraints.
- **Threat Definition:** A threat is defined as any action within the partial plan that has the potential to interfere with, or undo, an established causal link.
- **Threat Resolution:** If a threat is identified, it must be resolved, typically by adding constraints to ensure the threatening action either occurs before the causal link is established or after the needed condition is consumed.
- **Termination:** The iterative process of choosing a flaw, resolving it (and any resulting threats), and adding constraints and causal links continues until no remaining flaws are present in the partial plan.

3. Illustrate the formulation of a classical planning problem using the Blocks World example. Specifically, define a non-trivial Initial State and Goal State. Then, demonstrate how a specific action, such as moving a block, would be represented using the STRIPS representation, including its Preconditions, Add list, and Delete list.

Classical planning problems must be formalized using specific components like Initial State, Goal State, and Actions, with languages like STRIPS providing a structural means to represent the cause and effect of those actions.

- **Classical Planning Context:** The Blocks World problem is a canonical example used in classical planning, involving arranging blocks on a table to achieve a specific configuration.
- **Initial State Definition:** The Initial State sets the starting condition of the world, describing the arrangement of all blocks before the agent takes any action.
- **Non-Trivial Initial State Example:** A defined Initial State might be: Block A on the table, Block B resting on Block A, and Block C on the table.
- **Goal State Definition:** The Goal State specifies the exact desired outcome or condition that the final plan sequence must achieve.
- **Non-Trivial Goal State Example:** A target Goal State might be: Block C resting on Block B, Block B resting on Block A, and Block A resting on the table.
- **STRIPS Action Definition:** STRIPS is a representation language that defines an action using a set of preconditions and a set of effects (changes).
- **Action Representation:** Consider the action: Move(A, B, C), which moves Block A from its current location B to the destination C.
- **Preconditions Requirement:** The Preconditions define the set of conditions that *must* be true before the action can occur, such as: On(A, B), Clear(A), and Clear(C).

- **Add List Definition:** The Add list details facts that become true immediately after the action is successfully performed, such as: On(A, C) and Clear(B).
- **Delete List Definition:** The Delete list details facts that become false after the action is performed, such as: On(A, B) (A is no longer on B) and Clear(C) (C is no longer clear, as A is now on it).

5 Marks Questions

4. What is planning in AI and why is it considered critical for autonomous systems and decision-making processes?

Planning in AI is a deliberate, reasoned process focused on determining the necessary sequence of steps to transform a current situation into a desired goal state.

- **Definition of Planning:** Planning in AI is formally defined as the process of devising a specific plan of action to achieve a goal or a set of required goals.
- **Reasoning Component:** It fundamentally involves deep reasoning about all future actions and predicting the consequences that those actions will produce in the environment.
- **Goal Orientation:** The process aims to identify and find an optimal sequence of actions that achieves the desired end state.
- **Criticality for Autonomy:** Planning is considered critical for autonomous systems because it provides the necessary foundation for decision-making processes.
- **Practical Example:** A concrete application demonstrating this criticality is a robot whose task is to navigate a complex warehouse environment to successfully pick up and deliver packages.

5. List the core assumptions that define classical planning. Based on these assumptions, explain the difference between classical planning and planning in non-deterministic domains.

Classical planning operates under a set of rigid, simplifying assumptions that differentiate it structurally and conceptually from real-world planning in uncertain domains.

- **Classical Assumption (Determinism):** Classical planning assumes that all actions are deterministic, meaning an action will always produce one single, predictable, certain outcome.
- **Classical Assumption (Static/Observable):** The environment is assumed to be fully observable by the agent and static, meaning the world only changes due to the agent's actions.
- **Non-Deterministic Difference (Action Effects):** In non-deterministic domains, the key difference is that actions often have multiple possible outcomes, making the results uncertain.
- **Non-Deterministic Difference (Environment):** Non-deterministic environments pose challenges such as partial observability and dynamics, where the environment changes independently of the agent.
- **Planning Goal Difference:** While classical planning seeks an optimal sequence of guaranteed actions, non-deterministic planning strives to create robust plans that can function successfully despite inherent uncertainty.

6. Define the components of a classical planning problem, including the Initial State, Goal State, Actions, Transition Model, and Action Costs.

A classical planning problem must be precisely defined by a set of foundational components that establish the boundaries and rules within which the planning agent must operate.

- **Initial State:** This component defines the starting condition of the world—the complete set of facts or propositions that are true at the beginning of the planning process.
- **Goal State:** This specifies the desired final configuration or outcome that the plan must ultimately achieve for the problem to be solved.
- **Actions:** This is the complete set of all possible operations that the agent is capable of taking, each defined by its preconditions and resulting effects.
- **Transition Model:** This mathematically describes the rules of the world, detailing exactly how the application of any given action changes the state of the world.
- **Action Costs:** This specifies the associated cost (e.g., time, energy, monetary expense) for executing each action, which informs the search for an optimal (least-cost) plan.

7. Distinguish between forward state-space search (progression) and backward state-space search (regression) in classical planning algorithms. Discuss why backward search might be more efficient than forward search when solving problems aimed at achieving a specific complex configuration.

State-space search involves navigating through the space of possible world states, using either a progression approach starting from the initial state or a regression approach starting from the goal.

- **Forward Search (Progression):** This approach starts with the initial state and moves towards the goal by systematically applying all applicable actions to generate successor states.
- **Backward Search (Regression):** This approach starts with the goal state and moves backward by identifying predecessor states—finding actions that could have resulted in the current state.
- **Stopping Criterion (Forward):** The forward search process terminates when the current state being explored is evaluated and found to match the defined goal state.
- **Stopping Criterion (Backward):** The backward search process terminates when the current state being explored is evaluated and found to match the defined initial state.
- **Efficiency Reason:** Backward search can be more efficient because it focuses on only those preconditions that are strictly relevant to achieving the specific goal configuration, potentially leading to a much smaller effective search space than forward search.

8. Explain the function of a heuristic in AI planning. Define and differentiate between an admissible heuristic and a consistent heuristic, providing an example for each.

A heuristic is a guiding function used in informed search algorithms like A* to estimate the cost remaining to reach the goal, significantly improving search efficiency.

- **Heuristic Function:** A heuristic is fundamentally a function that estimates the cost of the path required to reach the goal state from the current state.
- **Purpose:** Heuristics serve to guide search algorithms, which effectively reduces the vast search space and leads to finding solutions much faster.

- **Admissible Heuristics:** A heuristic is defined as admissible if it guarantees that the estimated cost of reaching the goal never exceeds (never overestimates) the true, minimum cost required.
- **Admissible Example:** In a grid world, the Manhattan distance is an admissible heuristic because it is a simplified distance measure that cannot be longer than the true path cost.
- **Consistent Heuristics:** A heuristic is consistent if the estimated cost difference between any two adjacent nodes is always less than or equal to the actual cost of moving between those nodes, ensuring smooth and reliable guidance.

9. Describe the structure of the layered graph used in the GraphPlan algorithm. Explain the benefits of GraphPlan, particularly regarding its handling of action dependencies and parallel actions.

GraphPlan is the main graph-based planning algorithm that transforms a planning problem into a specialized layered structure to analyze action relationships and feasibility.

- **Layered Structure:** The planning graph consists of alternating levels, where one layer holds proposition nodes (representing facts) and the subsequent layer holds action nodes (representing operations).
- **Construction Process:** The graph is constructed by starting with the initial state propositions and iteratively adding applicable actions and the resulting propositions until the goal is reachable or the graph stabilizes.
- **Mutex Relations:** A critical structural element is the Mutex (mutual exclusion) relation, which is used to denote that two actions or two propositions cannot be simultaneously true or executed at the same level.
- **Benefit: Parallel Actions:** A primary advantage of GraphPlan is its ability to efficiently handle parallel actions, identifying which actions can occur concurrently without interference.
- **Benefit: Plan Existence:** The graph structure allows the algorithm to quickly and efficiently determine if a valid solution plan exists for the given problem.

10. What are the challenges inherent in non-deterministic planning? Explain the approach of Probabilistic Planning, noting which mathematical models (like MDPs) are often used and the planning goal.

Non-deterministic domains pose significant challenges because real-world environments introduce uncertainty that violates the core assumptions of classical planning.

- **Challenge 1: Uncertain Effects:** A key difficulty is that actions in these environments may produce multiple possible and unpredictable outcomes.
- **Challenge 2: Dynamic Environments:** The world itself may change independently of the agent's actions, complicating prediction and sequencing.
- **Probabilistic Planning Approach:** This planning methodology addresses uncertainty by explicitly assigning probabilities to the various possible action outcomes and world states.
- **Mathematical Models:** Probabilistic Planning frequently utilizes mathematical frameworks like Markov Decision Processes (MDPs) or Partially Observable MDPs (POMDPs).
- **Planning Goal:** The ultimate objective of Probabilistic Planning is not to find a single optimal sequence, but to find a policy or plan that maximizes the expected utility or minimizes the expected cost over time.

11. Compare and contrast the advantages and disadvantages of Reactive Planning and Replanning as approaches for dealing with uncertainty in non-deterministic environments.

Reactive Planning and Replanning are strategies for handling uncertainty, distinguished by their preference for immediate, local decisions versus planned adaptation.

- **Replanning Mechanism:** Replanning involves executing an initial plan and only generating a completely new plan when a failure or unexpected event is encountered during execution.
- **Reactive Mechanism:** Reactive planning avoids pre-calculating full plans, instead making immediate decisions based solely on the current observations (percepts) using condition-action rules.
- **Replanning Advantage/Disadvantage:** Replanning is highly adaptable to changing environments but its disadvantage is that frequent changes may make it computationally expensive due to the need for repeated planning.
- **Reactive Advantage:** Reactive planning offers very fast response times because it relies on simple rules, and it has low memory requirements as it does not store large plans.
- **Reactive Disadvantage:** A major drawback of Reactive Planning is that, due to its localized focus, it may not be able to find truly optimal solutions for complex, multi-step problems.

12. How does temporal planning differ from classical planning? Identify the characteristics of real-world actions (such as duration, overlap, and resources) that temporal planning must address.

Temporal planning, focused on time, schedules, and resources, provides the necessary complexity to handle real-world actions that violate the instantaneous assumption of classical planning.

- **Classical Simplification:** Classical planning simplifies the world by assuming that all actions happen instantaneously (taking zero time to complete).
- **Temporal Action Duration:** Temporal planning addresses the reality that real-world actions require an explicit, non-zero duration to complete (durative actions).
- **Action Overlap:** It must handle the fact that multiple actions can and often do occur simultaneously or overlap in time.
- **Temporal Constraints:** Temporal planning manages explicit relationships between actions, such as "Action A must complete *before* Action B starts" or "Action A must occur *during* Action B".
- **Resource Requirements:** Temporal planning incorporates resource management, addressing that actions may require specific resources (e.g., consumable fuel or reusable machinery) which must be scheduled and allocated.

13. Describe the three main types of resources relevant to scheduling and resource management in planning (Consumable, Reusable, Renewable), providing a brief description of each.

Resource management is a key component of temporal planning and scheduling, categorizing resources based on their availability after being used.

- **Consumable Resources:** These are resources that are entirely used up or depleted when they are consumed during the execution of a task (e.g., fuel or raw materials).
- **Reusable Resources:** These resources are made available again for subsequent use immediately after the action they were required for has been completed (e.g., machines, robots, or specific tools).

- **Renewable Resources:** These resources are replenished or restored over a period of time, meaning their supply regenerates automatically (e.g., a battery charge or available energy).
- **Resource Constraints:** Effective resource management must account for constraints such as capacity limitations and predefined availability windows for all resource types.
- **Management Goal:** The goal of management is to balance resource allocation and ensure resource utilization, potentially accounting for setup and cleanup time.

2 Marks Questions

14. List the three main categories of planning algorithms used in classical planning.

Classical planning algorithms are generally grouped based on the structure they search (states, plans, or graphs).

- **Category 1:** State-space search.
- **Category 2:** Plan-space search.

15. Name three common representation languages used to formalize planning problems.

To formalize the components of a planning problem, specific computational languages are utilized.

- **Language 1:** STRIPS (Stanford Research Institute Problem Solver).
- **Language 2:** PDDL (Planning Domain Definition Language).

16. What are the two core elements used in the STRIPS representation to define an action?

The STRIPS representation language defines the effects of an action through what is required before and what changes after execution.

- **Element 1:** Preconditions (conditions that must be true before the action can be applied).
- **Element 2:** Effects (the changes defined by the Add list and Delete list that occur after the action).

17. What is the definition of a heuristic in AI planning?

A heuristic is a guiding function designed to estimate remaining costs in a search space.

- **Definition:** A heuristic is a function that provides an estimate of the cost required to reach the goal state from a specific given state.
- **Function:** They are used to guide search algorithms, which helps to effectively reduce the size of the search space.

18. State two key concepts related to partial plans in Plan-Space Search (POP).

Partial-Order Planning uses specific relational concepts to maintain the integrity and connectivity of a partial plan.

- **Concept 1:** A causal link, which illustrates how one action fulfills the necessary precondition of another action.
- **Concept 2:** A threat, defined as an action that could potentially interfere with (or violate) an existing causal link.

19. Which heuristic is equivalent to performing uninformed search?

An uninformed search strategy corresponds to a heuristic that provides zero guidance to the algorithm.

- **Heuristic:** The Zero Heuristic (where $h = 0$).
- **Functionality:** This heuristic results in an uninformed search because it expands all possible paths equally, providing no cost estimation or preference.

20. In GraphPlan, what relationship is used to denote that two nodes are mutually exclusive?

GraphPlan uses a specific type of constraint to indicate that certain propositions or actions cannot co-exist in a single state or step.

- **Relationship:** The Mutex (mutual exclusion) relation.
- **Location:** These relations are used between action nodes and/or proposition nodes within the layered graph structure.

21. What are the four main approaches to non-deterministic planning?

Four distinct methodologies are recognized for planning and acting in domains characterized by uncertainty.

- **Approach 1 & 2:** Contingency Planning and Replanning.
- **Approach 3 & 4:** Reactive Planning and Probabilistic Planning.

22. List the three essential components that define a durative action.

Durative actions, which take time to complete, are defined by conditions spanning their start, middle, and end phases.

- **Component 1:** Start conditions and effects (what is true at the moment the action begins).
- **Component 2:** Invariant conditions (conditions that must hold true throughout the entire execution period).

23. In planning, what is the primary advantage of using GraphPlan over traditional state-space search?

GraphPlan offers efficiency gains, particularly in how it handles concurrent operations, due to its graphical representation.

- **Advantage:** The primary advantage is that GraphPlan can efficiently handle actions that occur in parallel.
- **Mechanism:** This efficiency is achieved because the layered graph structure allows for the analysis of action dependencies and non-interference (mutex relations).